

STUDENT MANAGEMENT SYSTEM

A CRUD-Based Web Application using Python Flask and SQLite

PROJECT REPORT

Submitted by

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Abstract

The Student Management System is a web-based application developed to manage student records efficiently using Create, Read, Update, and Delete (CRUD) operations. The application is developed using Python Flask as the backend framework, HTML and CSS for the frontend interface, and SQLite for database management.

The system allows users to add new student records, view existing records, update student information, and delete records when required. It provides a simple and user-friendly interface for managing student data.

The project demonstrates the development of a CRUD-based web application, database integration, frontend and backend communication, input handling, and deployment using Render. The application is also maintained in a GitHub repository for version control and project documentation.

1. Introduction

1.1 Overview of the Project

Student management is an important activity in educational institutions. Maintaining student information manually can be time-consuming and may lead to errors or difficulty in updating records.

The Student Management System is designed to provide a simple digital platform for managing student information. The application enables users to perform CRUD operations through a web interface.

The system uses Flask to handle application requests, SQLite to store student records, and HTML and CSS to create the user interface.

1.2 Problem Statement

Traditional student record management methods may involve manual data entry and maintenance. These methods can make it difficult to update information, retrieve records quickly, and maintain organized student data.

The proposed system addresses these challenges by providing a centralized web application for managing student records.

1.3 Objectives

- To develop a web-based Student Management System.
- To implement Create, Read, Update, and Delete operations.
- To integrate SQLite database with Flask.
- To provide a simple and user-friendly interface.
- To reduce manual effort in student record management.
- To deploy the application online using Render.
- To maintain the project source code using GitHub.

1.4 Scope of the Project

The system focuses on managing student records through a web application.

The main scope includes:

- Adding student records.
 - Viewing student records.
 - Editing student information.
 - Deleting student records.
 - Storing records in SQLite.
 - Deploying the application online.
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2. Requirement Specification

2.1 Functional Requirements

Requirement	Description
Create	Add a new student record
Read	Display existing student records
Update	Modify student information
Delete	Remove a student record
Database	Store student information
Web Interface	Provide an interface for users

2.2 Non-Functional Requirements

- Simple and user-friendly interface.
- Reliable database storage.
- Easy maintenance.
- Responsive layout where implemented.
- Proper handling of user input.
- Accessible through a web browser.

2.3 Hardware Requirements

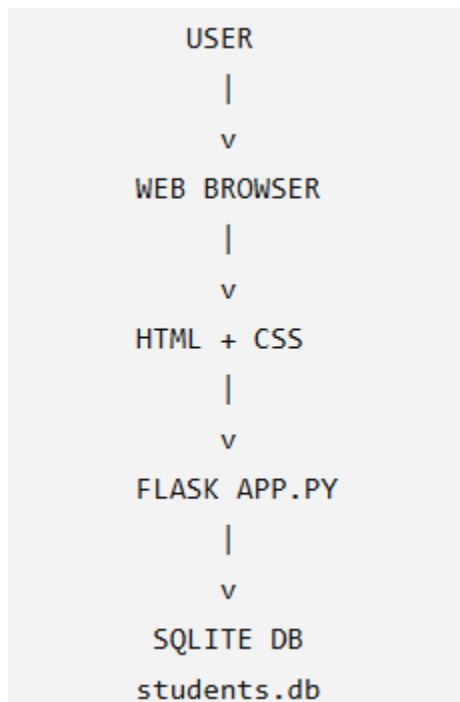
- Computer or laptop.
- Minimum 4 GB RAM.
- Internet connection for deployment.
- Keyboard and mouse.

2.4 Software Requirements

- Operating System: Windows.
 - Programming Language: Python.
 - Backend Framework: Flask.
 - Frontend: HTML, CSS.
 - Database: SQLite.
 - Code Editor: Visual Studio Code.
 - Version Control: GitHub.
 - Deployment Platform: Render.
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3. System Design

3.1 System Architecture



4 – Implementation

4.1 Project Structure

```
StudentManagementSystem/  
|  
├── static/  
│   └── style.css  
|  
├── templates/  
│   ├── index.html  
│   ├── add.html  
│   └── edit.html  
|  
├── app.py  
├── requirements.txt  
├── students.db  
└── README.md
```

4.2 Frontend Implementation

The frontend is developed using HTML and CSS. It provides the interface for entering student details, viewing records, and performing CRUD operations.

4.3 Backend Implementation

The backend is developed using Python Flask. It handles requests from the frontend and performs operations on the SQLite database.

4.4 Database Integration

SQLite is used to store student records. The Flask application connects to the database and performs operations such as inserting, retrieving, updating, and deleting records.

5 .Testing

5.1 Test Cases

Test Case	Operation	Expected Result
TC01	Add student	Student added
TC02	View students	Records displayed
TC03	Edit student	Details updated
TC04	Delete student	Record removed
TC05	Empty input	Validation/error handled

Results

The Student Management System was successfully developed and deployed as a web application.

The system supports student record management through CRUD operations. The application is available online through Render.

Live Website:

<https://studentmanagementsystem-kzsf.onrender.com/>

GitHub Repository:

<https://github.com/narmathavishali/StudentManagementSystem>

Conclusion

The Student Management System demonstrates the development of a CRUD-based web application using Python Flask and SQLite.

The project provides a simple solution for managing student records and demonstrates frontend development, backend processing, database integration, testing, version control, and deployment.

The application can be further enhanced with additional features such as authentication, search, filtering, and improved validation.

Future Enhancements

- User authentication and login.
 - Student search functionality.
 - Filtering student records.
 - Improved input validation.
 - Additional student information.
 - Enhanced user interface.
 - Advanced reporting features.
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11. References

1. Python Documentation
<https://docs.python.org/>
2. Flask Documentation
<https://flask.palletsprojects.com/>
3. SQLite Documentation
<https://www.sqlite.org/docs.html>
4. GitHub Documentation
<https://docs.github.com/>
5. Render Documentation
<https://render.com/docs>

